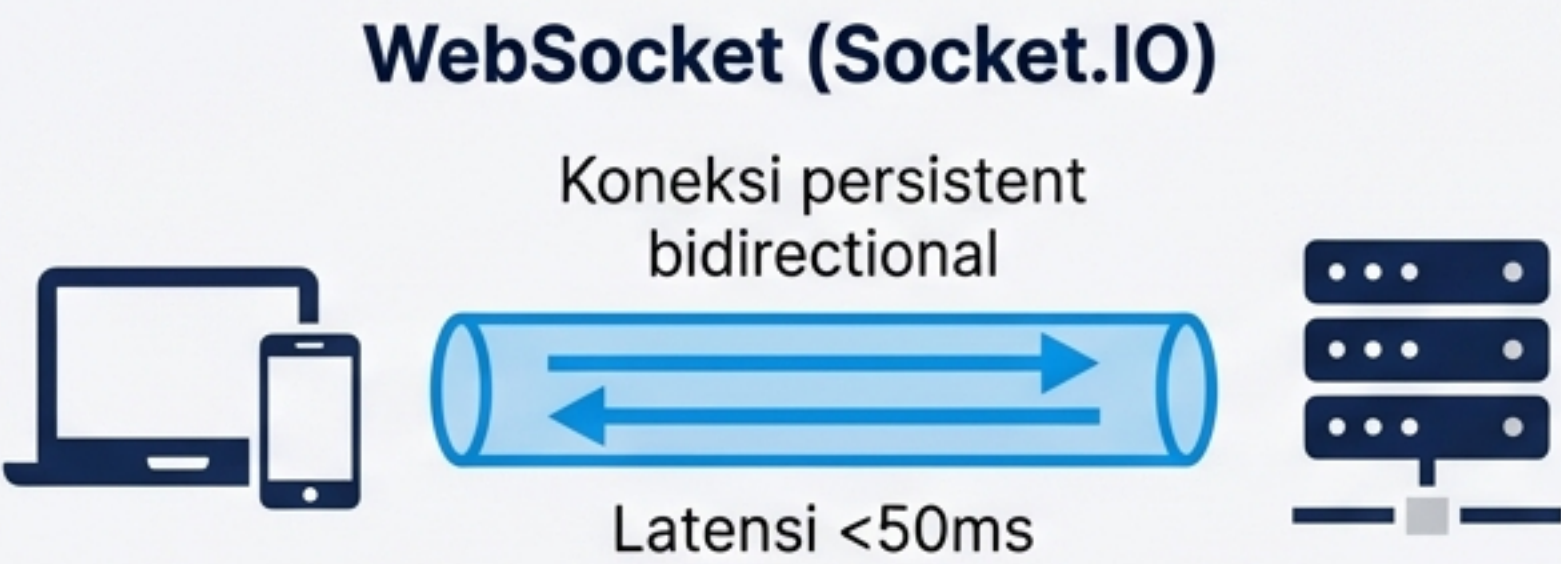
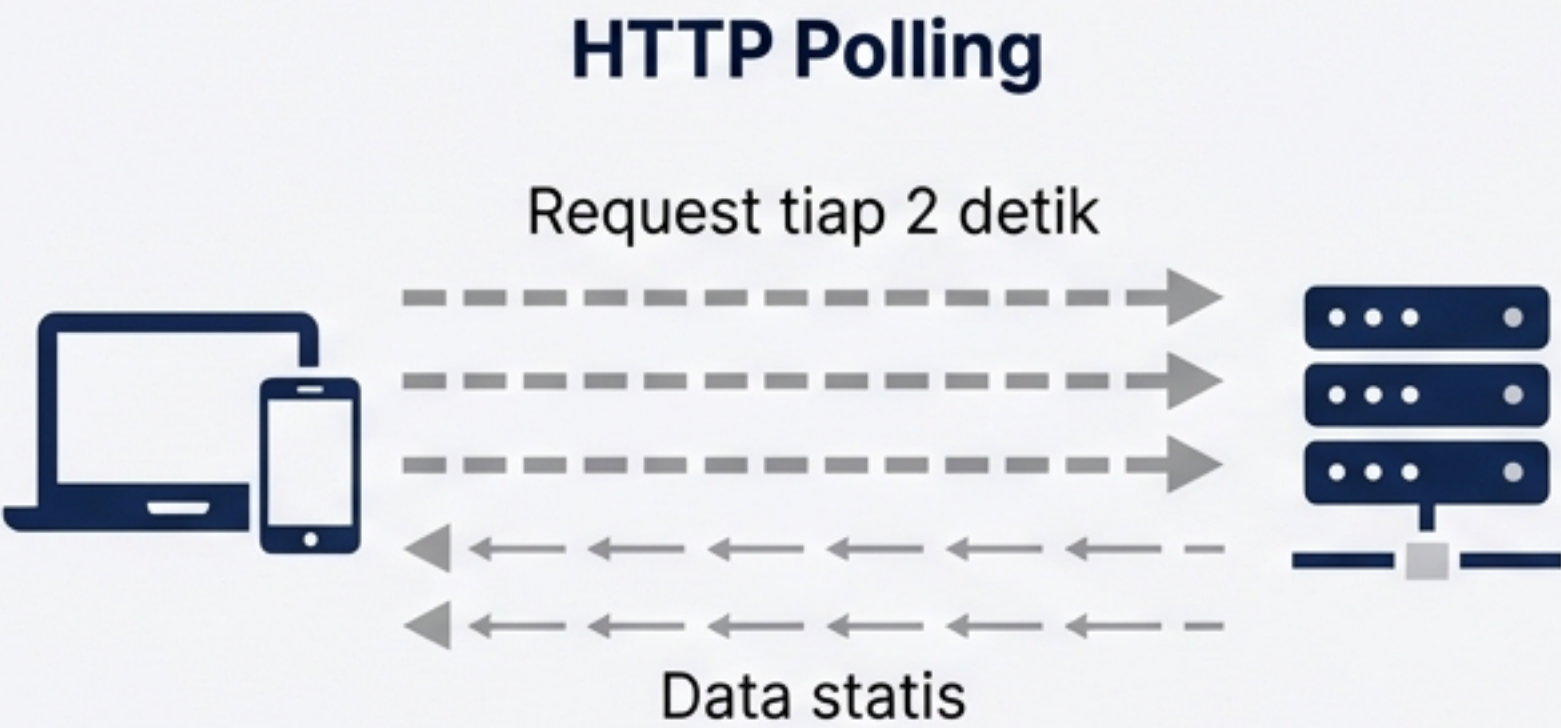




Upgrade 1: Dari Polling ke Koneksi Real-Time



Aspek	HTTP Polling	WebSocket
Koneksi	Request berulang tiap N detik	Persistent connection
Latensi	Tinggi (delay polling)	Sangat rendah (<50ms)
Bandwidth	Boros (header HTTP berulang)	Efisien (frame data kecil)
Use Case	Dashboard lambat	Chat, live update, game

```
npm install socket.io socket.io-client
```


Koreografi Kode Socket.IO: Server & Client

Node.js Server (server.js)

```
const { Server } = require('socket.io');  
  
io.on('connection', socket => {  
  io.emit('event'); // Broadcast ke semua  
  socket.to(room).emit('event');  
  // Spesifik room  
});
```

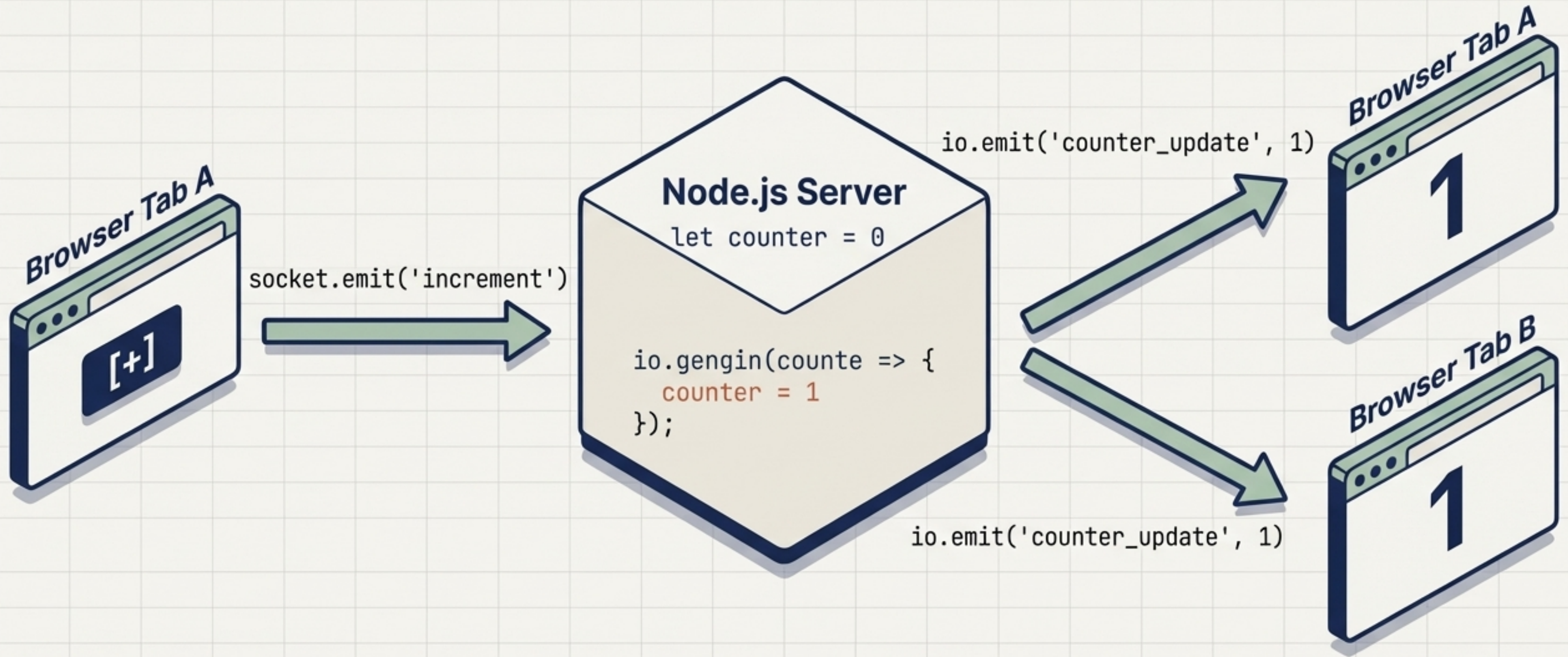
Browser / React (App.jsx)

```
import { io } from 'socket.io-client';  
  
useEffect(() => {  
  socket.on('event', handler);  
  return () => socket.off('event');  
}, []);
```

Perhatian: Gunakan `http.createServer(app)` agar Express dan Socket.IO berbagi port yang sama.

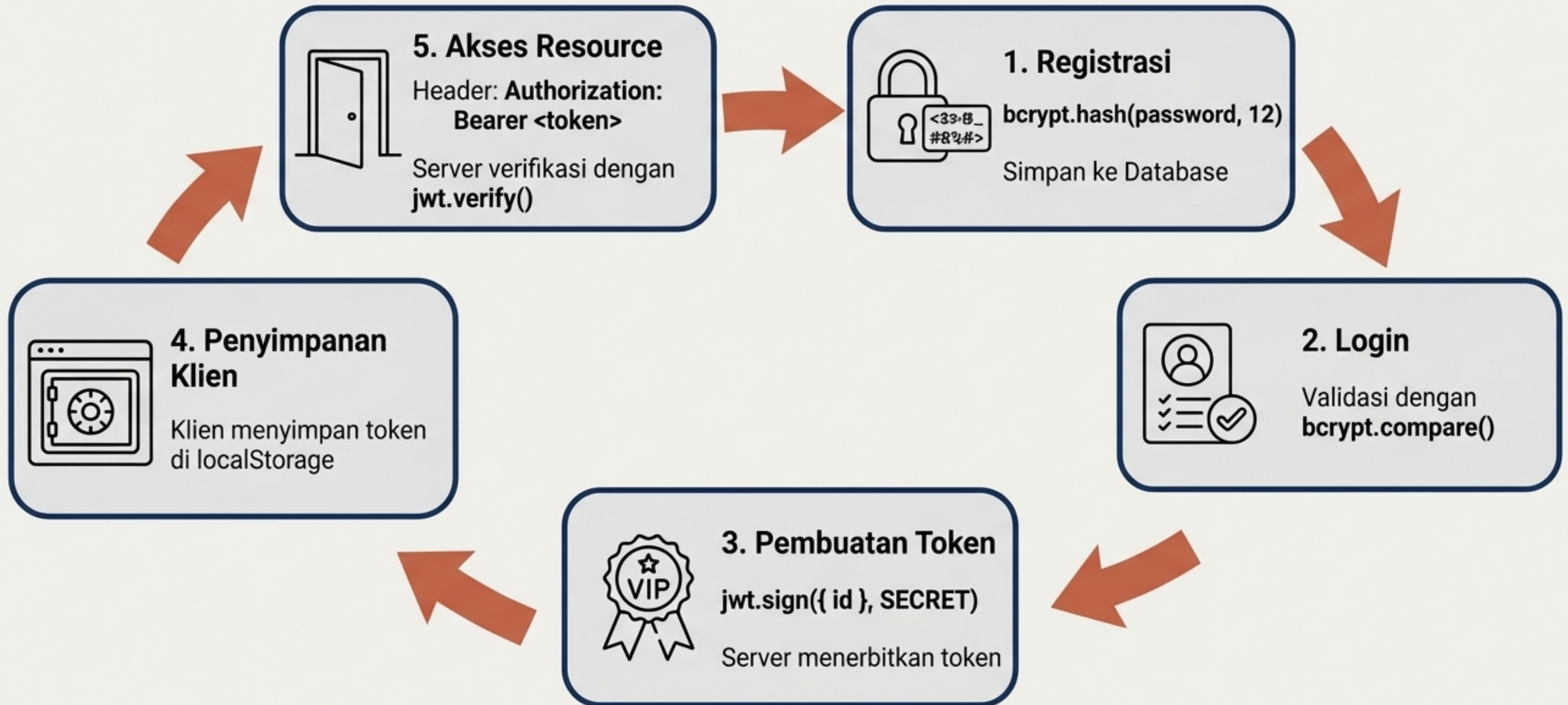
Penting: Cleanup (`socket.off`) saat unmount wajib dilakukan untuk mencegah memory leak!

Percobaan 5: Sinkronisasi Counter Real-Time

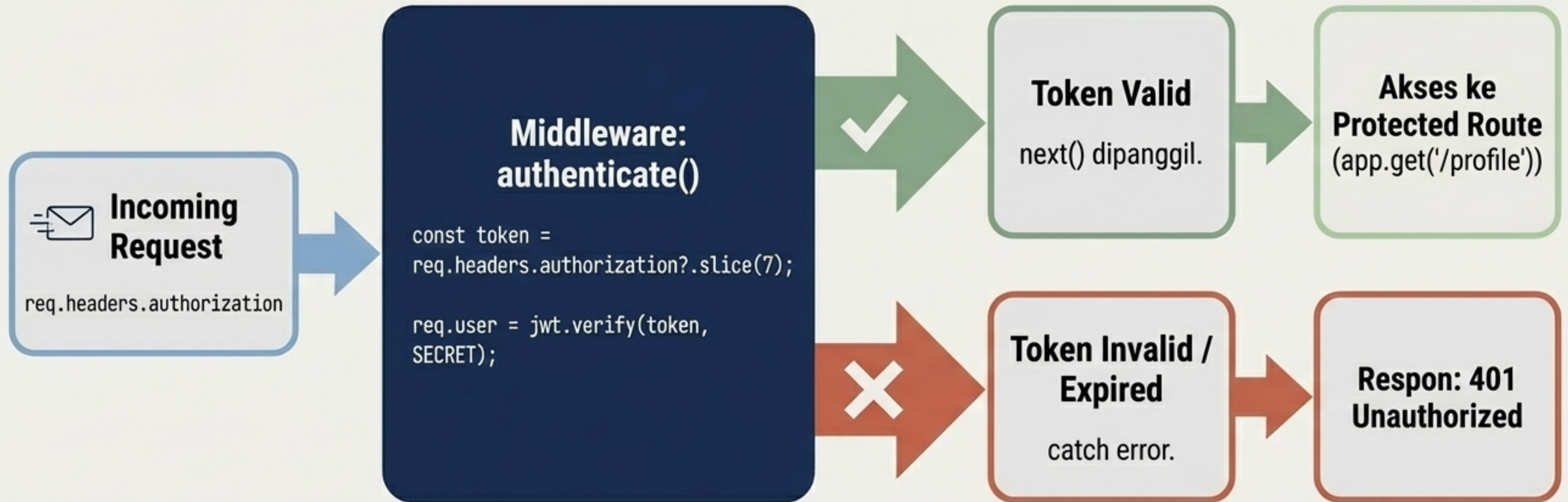


Kesimpulan: Dua tab browser tersinkronisasi tanpa refresh. Event di satu klien memicu broadcast instan ke seluruh klien.

Upgrade 2: Autentikasi Stateless dengan JWT



Middleware Auth: Sang Penjaga Gerbang API



Peringatan Keamanan: Jangan pernah menyimpan password dalam format plain text!
Selalu gunakan bcrypt.

Upgrade 3: Arsenal Database Modern

MySQL (mysql2)	PostgreSQL (pg)	MongoDB (Mongoose)
Relational (SQL) • Query Param: Menggunakan ? • Contoh: <code>pool.execute('...', [param])</code> • Kelebihan: <code>createPool</code> efisien, stabil, tradisional. • Use Case: Aplikasi web standar, CMS dasar.	Relational + Native JSON • Query Param: Menggunakan \$1, \$2 • Contoh: <code>pool.query('...', [param])</code> • Kelebihan: Tipe kolom JSONB, full-text search, fungsi agregasi. • Use Case: IoT Dashboard, E-Commerce, Production.	Document-Centric (NoSQL) • Struktur: Schema → Model → CRUD • Fitur: <code>populate()</code> untuk join data. • Kelebihan: Schema fleksibel, iterasi cepat. • Use Case: Blog, Kanban Board, AI Chatbot.

Prisma ORM: Jembatan Tipe-Aman Multi-Database

schema.prisma

```
model Todo {  
  id      Int      @id @default(autoincrement())  
  title   String  
  done    Boolean @default(false)  
}
```

Type-Safe:

`prisma.todo.findMany()`
menjamin struktur data sesuai skema.

Prisma Engine

`npx prisma
migrate dev`

Agnostik: Ganti database tanpa menulis ulang logika SQL.



SQLite



MySQL

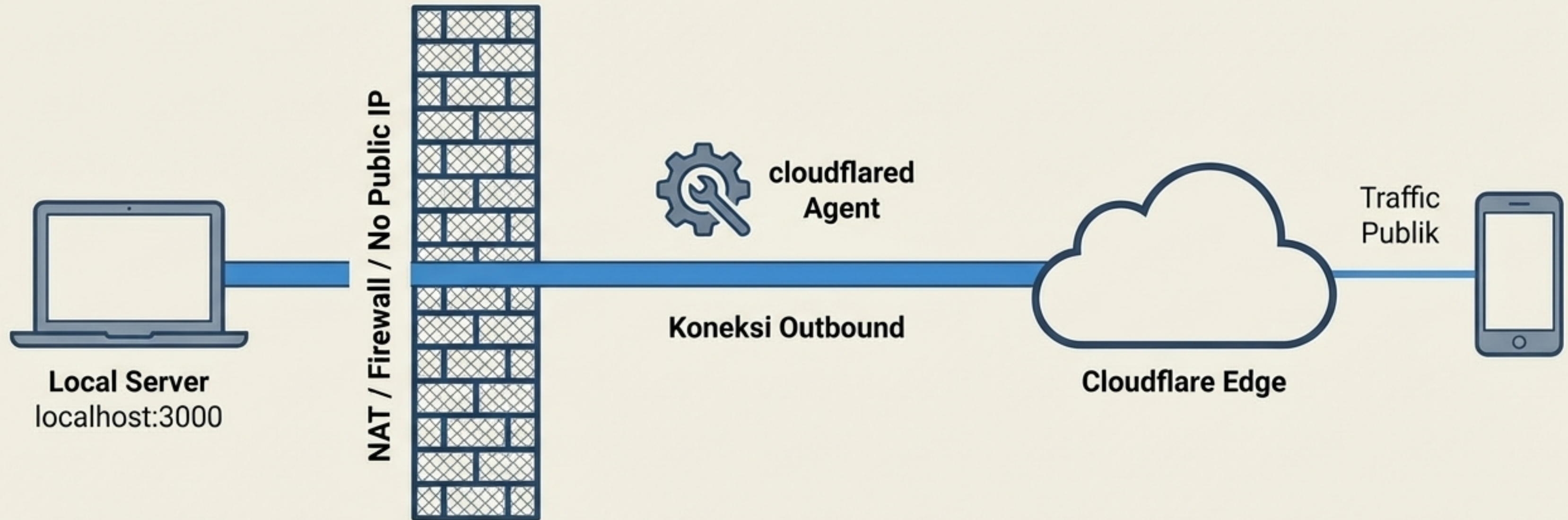


PostgreSQL



MongoDB

Upgrade 4: Menembus Tembok NAT (Cloudflare Tunnel)



Keuntungan: Tidak perlu IP Publik. Tidak perlu Port Forwarding. Tidak perlu merubah aturan Firewall. Aman dan instan.

Percobaan 6: Localhost Mengudara

1. Eksekusi Terminal

Terminal

```
> winget install Cloudflare.cloudflared
> cloudflared tunnel --url http://localhost:3000
...
INF +-----+
INF | https://random-word.trycloudflare.com |
INF +-----+
```

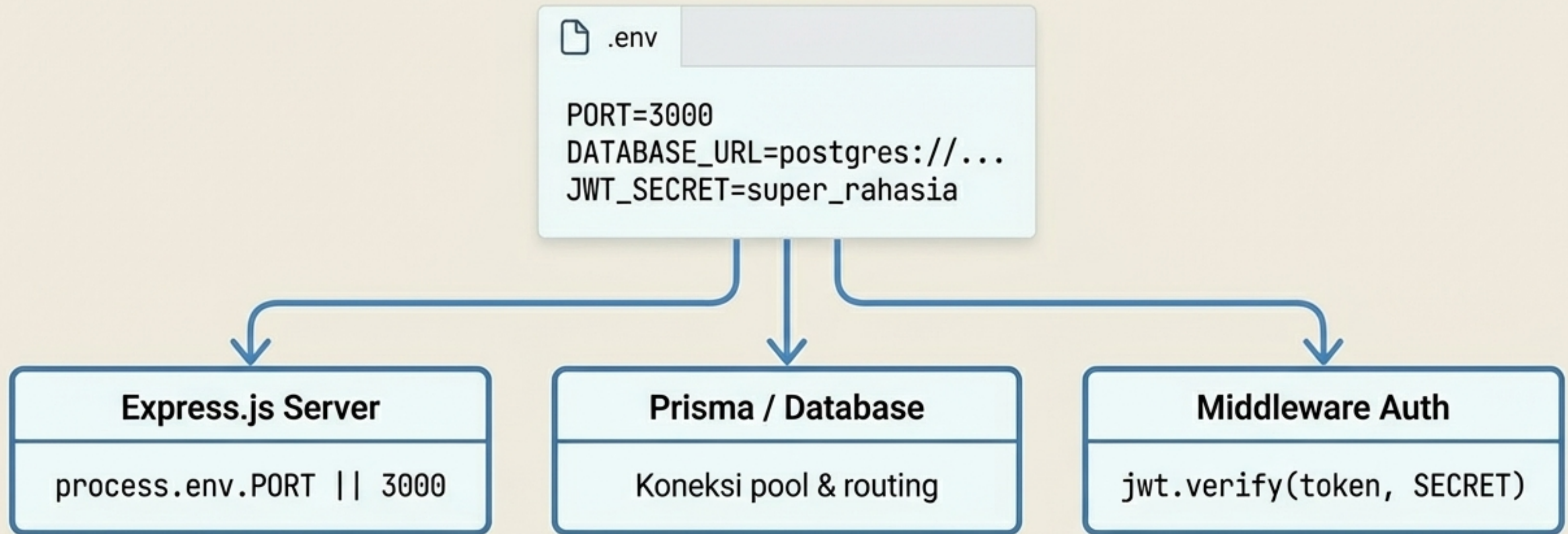
2. Akses Global

<https://random-word.trycloudflare.com/api/todos>

4G / Data Seluler (Bukan WiFi)

```
[{
  id: 1,
  title: 'Test Tunnel',
  done: false
}]
```

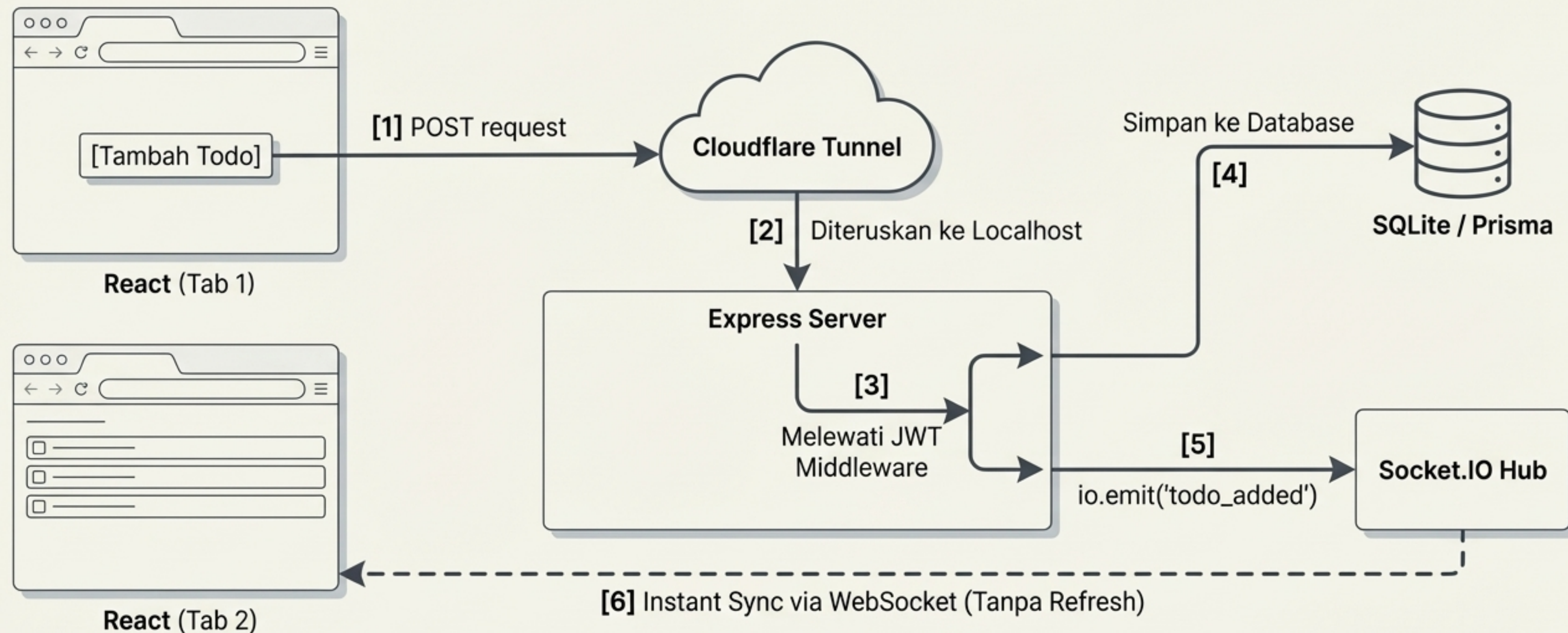

Kunci Ekosistem: Environment Variables



Aturan Emas: File .gitignore WAJIB menyertakan .env!

Jangan pernah hardcode credential langsung di dalam source code.

Sintesis Final: Arsitektur Ekosistem Full-Stack (Percobaan 7)



Kombinasi Express, Database, React, dan Socket.IO—diekspos via Cloudflare—menciptakan aplikasi berskala produksi yang real-time dan dapat diakses publik.